P.6 SCIENCE

LESSON PLANS

TERM 1 2026

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PRIMARY SIX SCIENCE LESSON PLANS TERM ONE 2026

LESSON PLAN 1: CLASSIFICATION OF ANIMALS (INTRODUCTION)

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Six	Integrated Science		60 Minutes

THEME	TOPIC	CONTENT
The World of Living Things	Classification of Animals	Introduction to classification; vertebrates and invertebrates

COMPETENCES:

- **Subject Competence:** The learner identifies biological features used to classify animals and distinguishes between vertebrates and invertebrates.
- **Language Competence:** The learner uses terms like classification, vertebrates, invertebrates, backbone, and characteristics correctly.

TEACHING METHODS/TECHNIQUES:

- Question and answer
- Demonstration
- Guided discovery
- Use of charts/pictures
- Discussion
- Group work
- Note-taking

LEARNING OUTCOMES:

By the end of this lesson, learners should be able to:

1. State the biological features used to classify animals
2. Give reasons why animals are classified
3. Define vertebrates and invertebrates
4. Identify characteristics common to all animals

TEACHING/LEARNING MATERIALS:

- Chart showing different animals (vertebrates and invertebrates)
- Pictures of animals with and without backbone (refer to source document images)
- Real specimens (if available: insect, small fish, snail)
- Chalkboard/whiteboard
- Learners' notebooks
- The P.6 Integrated Science Lesson Notes (Source document)

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	Greets learners warmly and asks about their previous knowledge of animals Shows pictures of different animals (cow, fish, grasshopper, snail) as shown on page 6 Asks: "What do you notice about these animals? How are they similar or different?" Listens to learners' responses Introduces the topic: Classification of Animals	Question and answer Brainstorming Use of pictures	Responds to greeting Observes the pictures carefully Mentions names of animals seen Shares what they know about animals Listens attentively
15 mins	**EXPLORE**	Divides learners into small groups Gives each group pictures or names of different animals Asks them to group the animals based on similarities they observe Guides groups to consider: body covering, how they move, where they live Lets groups present their findings to the class Records their groupings on the board	Group work Guided discovery Exploration Discussion	Works cooperatively in groups Observes animal pictures Discusses similarities and differences Groups animals according to observed features Presents findings to class Listens to other groups' presentations
20 mins	**EXPLAIN**	Explains what classification means: grouping things based on similarities States the biological features used to classify animals (page 6): • The mode of feeding • The mode of reproduction • The body structure/shape • Mode of fertilisation Gives reasons why animals are classified (page 6): • For easy identification of animals • For easy study of animals • For easy naming of animals • For easy understanding of animals Defines vertebrates: animals that have a backbone Defines invertebrates: animals that have no backbone States characteristics common to all animals (page 6): • Animals are made up of many cells	Lecture method Explanation Demonstration Use of chart Note-taking	Listens attentively to teacher Observes the classification table Writes notes in their books Answers questions asked Asks for clarification where needed Identifies animals as vertebrates or invertebrates

		• Animals feed on ready-made food • Animals lack chlorophyll • Animals have natural defensive mechanisms Draws the classification table on the board (page 6): ANIMALS → Vertebrates → Warm blooded (Mammals, Birds) ANIMALS → Vertebrates → Cold blooded (Reptiles, Fish, Amphibians) ANIMALS → Invertebrates → Arthropods, Molluscs, Worms Refers to the diagram on page 6 showing classification of vertebrates and invertebrates Asks learners to copy the notes		
10 mins	**ELABORATE**	Shows learners pictures of different animals and asks them to identify whether they are vertebrates or invertebrates Gives examples: • Fish (vertebrate) • Grasshopper (invertebrate) • Frog (vertebrate) • Snail (invertebrate) Asks: "Why is a cow classified as a vertebrate?" Guides learners to give more examples from their local environment Asks learners to complete the table in the source material (page 7, question 11)	Guided practice Application Question and answer Use of examples	Identifies vertebrates and invertebrates from pictures Gives reasons for their answers Provides examples from local environment Completes the classification table Participates actively in discussion
5 mins	**EVALUATE**	Asks oral questions to check understanding: • "What is classification?" • "Mention two biological features used to classify animals" • "What is a vertebrate?" • "Give two examples of invertebrates" Gives class work from the source material (page 7, questions 1-5)	Oral assessment Question and answer Written assignment	Answers oral questions confidently Writes class work in their books Shows understanding of the topic
5 mins	**EXTEND**	Tells learners to observe animals around their home and list five vertebrates and five invertebrates they see Asks them to find out the five groups of vertebrates for the next lesson	Home assignment Preview next lesson	Listens to home assignment Asks questions where unclear Prepares for next lesson

		Prepares learners for the next topic: groups of vertebrates Encourages them to read pages 6-7 of their source material at home		

CLASS WORK:

1. Mention any two biological features used to classify animals.
2. Give any one characteristic common to all animals.
3. Why is a bat referred to as a warm-blooded vertebrate?
4. Write down any two reasons why animals are called living things.
5. List any two ways in which animals are different from plants.

LESSON EVALUATION:

STRENGTH:	**AREAS WHICH NEED IMPROVEMENT:**	**WAY FORWARD:**

LESSON PLAN 2: MAMMALS

DATE	**CLASS**	**SUBJECT / L. AREA**	**NO. OF CHILDREN**	**TIME**
	Primary Six	Integrated Science		60 Minutes

THEME	**TOPIC**	**CONTENT**
The World of Living Things	Classification of Animals	Mammals: characteristics and groups

COMPETENCES:

- **Subject Competence:** The learner identifies mammals, states their characteristics, and groups them into different categories.
- **Language Competence:** The learner uses terms like mammals, primates, herbivores, carnivores, omnivores, and marsupials correctly.

TEACHING METHODS/TECHNIQUES:

- Question and answer
- Demonstration
- Guided discovery
- Use of charts/pictures
- Discussion
- Group work

- Note-taking

LEARNING OUTCOMES:

By the end of this lesson, learners should be able to:

1. Identify mammals as a group of vertebrates
2. State the general characteristics of mammals
3. Name the different groups of mammals with examples
4. Give examples of mammals from each group

TEACHING/LEARNING MATERIALS:

- Chart showing different mammals (refer to images on pages 8, 11, 12)
- Pictures of primates, sea mammals, insect-eating mammals, flying mammals, pouched mammals, egg-laying mammals
- Chalkboard/whiteboard
- Learners' notebooks
- The P.6 Integrated Science Lesson Notes (Source document)

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	Greets learners and reviews the previous lesson on vertebrates and invertebrates Asks: "What are vertebrates?" "Name the five groups of vertebrates" (MR FAB: Mammals, Reptiles, Fish, Amphibians, Birds) Shows a picture of a cow, a dog, and a human (refer to image on page 8) Asks: "What do these animals have in common?" Introduces the topic: Mammals	Question and answer Review Use of pictures Brainstorming	Responds to greeting Answers review questions Observes the pictures Mentions what the animals have in common (body hair, feed young on milk) Listens attentively
15 mins	**EXPLORE**	Divides learners into groups Gives each group pictures of different mammals (monkey, whale, bat, kangaroo, hedgehog, platypus) from pages 8, 11, 12 Asks them to observe the animals and note their differences Guides groups to discuss: "Where do these animals live?" "How do they move?" "What do they eat?" "How do they reproduce?" Lets each group present their	Group work Guided discovery Observation Discussion	Works cooperatively in groups Observes pictures carefully Discusses differences among mammals Notes where each animal lives and how it moves Presents findings to

		observations Records their findings on the board		class Listens to other groups
20 mins	**EXPLAIN**	Defines mammals: warm blooded vertebrates with mammary glands which produce milk (page 8) States characteristics of mammals (page 8): • They have mammary glands that produce milk for their young ones • They have a hairy body (hair insulates against heat loss) • Most mammals produce live young ones Explains the different groups of mammals with examples (pages 8-12): **1. Primates** (most advanced with well-developed brain; omnivores) Examples: monkey, gorilla, chimpanzee, baboon, human being **2. Flesh-eating mammals** (carnivores with well-developed canine teeth) Examples: lion, leopard, hyena, cheetah, wolf, jackal **3. Sea/marine mammals** (live in water; have blubber; flippers) Examples: whale, dolphin, seal, otter, porpoise, dugong, polar bear (refer to page 11 images) **4. Insect-eating mammals** (feed on insects; strong claws; long sticky tongue) Examples: hedgehog, mole, pangolin, aardvark, elephant shrew (refer to page 11 images) **5. Flying mammals** (limbs modified into wings; chiroptera) Example: bat (page 12) **6. Pouched mammals** (have pouch for carrying joeys; marsupials) Examples: kangaroo, wombat, koala bear, wallaby, opossum (refer to page 12 images) **7. Egg-laying mammals** (reproduce by laying eggs; monotremes) Examples: duck-billed platypus, spiny anteater (refer to page 12 images) Explains why a bat is a mammal: has mammary glands, hairy body,	Lecture method Explanation Use of pictures Demonstration Note-taking	Listens attentively Observes pictures of different mammal groups Identifies characteristics of each group Writes notes in their books Answers questions Asks for clarification

		produces live young ones (page 12) Notes that a bat is nocturnal and uses echoes to locate food Refers to diagrams on pages 8, 11, 12 Asks learners to copy the notes		
10 mins	**ELABORATE**	Shows pictures of different mammals and asks learners to identify the group they belong to Gives examples: • Seal (sea mammal) • Pangolin (insect-eating mammal) • Bat (flying mammal) • Kangaroo (pouched mammal) • Platypus (egg-laying mammal) Asks: "Why is a whale not a fish but a mammal?" Guides learners to answer questions from the source material (page 10, questions 1-7 and page 13, questions 1-6) Discusses the importance of mammals in the environment	Guided practice Question and answer Application Discussion	Identifies groups of mammals from pictures Gives reasons for their answers Answers questions from the source material Participates in discussion Writes answers in their books
5 mins	**EVALUATE**	Asks oral questions: • "What are mammals?" • "Name three characteristics of mammals" • "To which group of mammals does a lion belong?" • "Why is a bat able to fly at night without hitting obstacles?" • "Name two egg-laying mammals" Gives class work from the source material (page 13, questions 1-6)	Oral assessment Question and answer Written assignment	Answers oral questions confidently Copies class work Shows understanding of the topic
5 mins	**EXTEND**	Tells learners to find out about birds for the next lesson Asks them to observe birds around their home and note their features Encourages them to read pages 13-14 of their source material Prepares them for the next topic: Birds	Home assignment Preview next lesson	Listens to home assignment Asks questions where unclear Prepares for next lesson

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CLASS WORK:

1. Name the group of mammals which have well developed canine teeth.
2. What type of fertilization do mammals undergo?

3. To which group of mammals does a gopher belong?
4. Give any one example of flesh-eating mammals.
5. Why do flesh-eating mammals have well developed canine teeth?
6. State any two other characteristics of flesh-eating mammals.
7. Name the layer of fats that keeps sea mammals warm when in cold water of an ocean.
8. Write down any two examples of sea mammals.

LESSON EVALUATION:

STRENGTH:	AREAS WHICH NEED IMPROVEMENT:	WAY FORWARD:

LESSON PLAN 3: BIRDS (CHARACTERISTICS AND EXTERNAL FEATURES)

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Six	Integrated Science		60 Minutes

THEME	TOPIC	CONTENT
The World of Living Things	Classification of Animals	Birds: characteristics, external features, and functions

COMPETENCES:

- **Subject Competence:** The learner identifies birds, states their characteristics, and explains the functions of their external features.
- **Language Competence:** The learner uses terms like feathers, beak, nictitating membrane, streamlined body, and hollow bones correctly.

TEACHING METHODS/TECHNIQUES:

- Question and answer
- Demonstration
- Guided discovery
- Use of charts/pictures
- Discussion
- Group work
- Note-taking

LEARNING OUTCOMES:
By the end of this lesson, learners should be able to:

1. State the characteristics of birds

2. Identify the external features of a bird
3. Explain the functions of different parts of a bird
4. Give examples of birds

TEACHING/LEARNING MATERIALS:

- Chart showing a bird with labeled parts (refer to image on page 14)
- Pictures of different birds (chicken, eagle, duck, ostrich)
- Real specimen (if available: feather, bird's nest)
- Chalkboard/whiteboard
- Learners' notebooks
- The P.6 Integrated Science Lesson Notes (Source document)

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	Greets learners and reviews mammals from previous lesson Asks: "What are the five groups of vertebrates?" (MR FAB) Shows a picture of a bird (chicken) from page 14 Asks: "What animal is this?" "How do you know it is a bird?" Listens to learners' responses Introduces the topic: Birds	Question and answer Review Use of pictures Brainstorming	Responds to greeting Answers review questions Observes the picture Mentions features that make it a bird Listens attentively
15 mins	**EXPLORE**	Divides learners into groups Gives each group a picture of a different bird (eagle, duck, chicken, ostrich) from pages 14-16 Asks them to observe and list all the features they see on the bird Guides them to identify: beak, feathers, wings, legs, claws, eyes, tail Asks: "What do you think each part is used for?" Lets groups present their findings Records their observations on the board	Group work Guided discovery Observation Discussion	Works cooperatively in groups Observes bird pictures carefully Lists external features of birds Discusses possible functions of each part Presents findings to class Listens to other groups
20 mins	**EXPLAIN**	States characteristics of birds (page 13): • They have feathers covering their bodies • They have scales on their legs • They have streamlined bodies (reduce	Lecture method Explanation Demonstration Drawing Note-taking	Listens attentively Observes the diagram Identifies each part as teacher explains

		friction in air during flight) • They have hollow bones (reduce weight in air during flight) • They have a nictitating membrane on their eyes (protects against strong wind) Draws and labels the external features of a bird on the board (as on page 14) Explains functions of each feature (pages 14-15): **Beak:** For picking and breaking food; for protection; for pecking trees to look for insects; for turning eggs during incubation **Feathers:** For flight; protect skin from injuries; keep the bird warm **Nostril:** For smelling; lets in air during breathing **Wings:** For flight **Claws:** For holding prey firmly; for perching; for climbing; for scratching ground; for protection **Tail:** For changing direction during flight **Eyes:** For seeing; have nictitating membrane **Earlobes:** For protecting the ears Refers to the diagram on page 14 Asks learners to copy the diagram and notes		Writes notes and draws the diagram Asks questions Participates in discussion
10 mins	**ELABORATE**	Asks learners to label parts of a bird on a diagram Gives questions: • "Why do birds have hollow bones?" • "How is the nictitating membrane useful?" • "Why do birds raise their feathers during cold weather?" Guides them to answer questions from the source material (page 17, questions 4, 6) Discusses reasons why birds are able to fly (page 21): • Forelimbs modified into wings • Flight feathers on their wings • Strong pectoral muscles	Guided practice Question and answer Application Discussion	Labels parts of a bird Answers questions correctly Writes answers in their books Participates in discussion Gives reasons for bird adaptations
5 mins	**EVALUATE**	Asks oral questions: • "Name three characteristics of birds" • "What is the function of feathers?" • "Why do birds have a streamlined	Oral assessment Question and answer Written assignment	Answers oral questions confidently Copies class work

		body?" • "What protects a bird's eyes during flight?" Gives class work from the source material (page 17, questions 1-6)		Shows understanding of the topic
5 mins	**EXTEND**	Tells learners to observe birds in their environment and note the different types of beaks and feet they have Asks them to find out about groups of birds for the next lesson Encourages them to read pages 14-16 of their source material Prepares them for next topic: Groups of birds	Home assignment Preview next lesson	Listens to home assignment Asks questions where unclear Prepares for next lesson

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CLASS WORK:

1. State any two characteristics of birds.
2. Name the feature on an eagle that enables it to tear flesh.
3. Apart from an eagle, name any two other examples of birds of prey.
4. State any two features used to classify birds.
5. Write down any two examples of scavenger birds.
6. In which way are scavenger birds useful in the environment?
7. Give any two uses of feathers to birds.

LESSON EVALUATION:

STRENGTH:	**AREAS WHICH NEED IMPROVEMENT:**	**WAY FORWARD:**

LESSON PLAN 4: GROUPS OF BIRDS

DATE	**CLASS**	**SUBJECT / L. AREA**	**NO. OF CHILDREN**	**TIME**
	Primary Six	Integrated Science		60 Minutes

THEME	**TOPIC**	**CONTENT**
The World of Living Things	Classification of Animals	Groups of birds based on feeding and movement

COMPETENCES:

- **Subject Competence:** The learner classifies birds into different groups based on their mode of feeding, beak shape, and feet structure.

- **Language Competence:** The learner uses terms like birds of prey, scavengers, perching birds, webbed feet, and conical beak correctly.

TEACHING METHODS/TECHNIQUES:

- Question and answer
- Demonstration
- Guided discovery
- Use of charts/pictures
- Discussion
- Group work
- Note-taking

LEARNING OUTCOMES:
By the end of this lesson, learners should be able to:

1. Identify factors used to classify birds
2. Name different groups of birds with examples
3. Match bird beaks and feet to their functions
4. State the importance and dangers of birds in the environment

TEACHING/LEARNING MATERIALS:

- Chart showing different bird beaks and feet (refer to images on pages 17-21)
- Pictures of birds of prey, scavengers, swimming birds, scratching birds, climbing birds, perching birds
- Chalkboard/whiteboard
- Learners' notebooks
- The P.6 Integrated Science Lesson Notes (Source document)

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	Greets learners and reviews characteristics of birds Asks: "What are the main features of birds?" "How do birds differ from each other?" Shows pictures of different birds (eagle, duck, chicken, parrot) from pages 15-16 Asks: "Look at their beaks and feet. Are they the same? Why are they different?" Introduces the topic: Groups of birds	Question and answer Review Use of pictures Brainstorming	Responds to greeting Answers review questions Observes pictures of different birds Notices differences in beaks and feet Listens attentively

15 mins	**EXPLORE**	Divides learners into groups Gives each group pictures showing different bird beaks and feet (refer to pages 17-19) Asks them to match the beak to the type of food the bird eats Guides them to discuss: • "Why does this bird have a hooked beak?" • "Why does this bird have webbed feet?" Lets groups present their findings Records their observations on the board	Group work Guided discovery Observation Discussion	Works cooperatively in groups Observes pictures of beaks and feet Discusses the relationship between beak shape and diet Matches beaks to food types Presents findings to class Listens to other groups
20 mins	**EXPLAIN**	Explains factors to consider when classifying birds (page 14): • Mode of feeding • Nature of the feet • Shape of the beak • Mode of movement Describes each group of birds with characteristics and examples (pages 14-21): **1. Birds of prey** (pages 14-15): • Feed on prey hunted and killed • Strong sharp hooked beak (tear flesh) • Strong sharp curved talons (grip prey) • Strong eyesight • Examples: eagle, kite, hawk, owl, falcon (refer to page 15 images) **2. Scavenger birds** (pages 15-16): • Feed on leftover/abandoned flesh • Strong hooked beak for tearing carcasses • Clean environment • Examples: marabou stork, crow, vulture (refer to page 16) **3. Swimming birds** (page 16): • Webbed feet for paddling • Spoon-shaped beak for sieving food from mud/water • Glands produce oil for waterproof skin • Examples: duck, swan, pelican, goose, penguin **4. Scratching birds** (page 18):	Lecture method Explanation Use of pictures Demonstration Note-taking	Listens attentively Observes pictures of each bird group Identifies characteristics of each group Writes notes in their books Answers questions Asks for clarification

		• Strong blunt claws for scratching ground • Strong short pointed beaks for picking food • Examples: chicken, turkey, guineafowl **5. Climbing birds** (pages 18-19): • Two toes forward, two backward for climbing trees • Strong pointed chisel-shaped beaks for digging holes • Examples: parrot, woodpecker (refer to page 19 images) **6. Perching birds** (pages 19-20): • Three toes forward, one backward for holding firmly on branches • Further grouped into: - Seed eaters: short strong conical beaks; examples: pigeon, dove, weaver bird - Insect eaters: short narrow beaks; examples: sparrow, robin, swallow (page 20) - Nectar suckers: long slender beaks; help in pollination; examples: sunbird, hummingbird (page 20) - Fruit eaters: long stout beaks; help in seed dispersal; example: hornbill (page 21) **7. Flightless birds** (page 21): • Small weak wings compared to body size; cannot fly • Examples: ostrich, emu, kiwi, penguin (refer to page 21 images) Explains importance of birds (page 21): • Food (meat, eggs) • Pollination • Pest control • Income • Cleaning environment Explains dangers of birds (page 21): • Spoil crops • Cause accidents at airports • Carry fleas • Kill chicks Refers to diagrams on pages 15-21 Asks learners to copy notes		

10 mins	**ELABORATE**	Shows pictures of bird beaks and feet and asks learners to identify the group and explain why Gives examples: • Webbed feet (swimming birds) • Hooked beak (birds of prey) • Strong blunt claws (scratching birds) • Two toes forward/backward (climbing birds) Asks: "How is the feeding of a pigeon different from that of a robin?" Guides learners to answer questions from the source material (pages 17-18, 22, questions 7-11) Discusses similarities between birds and mammals (page 21): • Both warm-blooded vertebrates • Both breathe through lungs • Both undergo internal fertilization • Both have streamlined bodies	Guided practice Question and answer Application Discussion	Identifies bird groups from beaks and feet Explains their reasoning Answers questions from source material Participates in discussion Writes answers in their books
5 mins	**EVALUATE**	Asks oral questions: • "Name three groups of birds" • "What type of beak does a bird of prey have?" • "Why does a duck have webbed feet?" • "How are sunbirds useful to plants?" • "Why is an ostrich unable to fly?" Gives class work from the source material (page 22, questions 1-9)	Oral assessment Question and answer Written assignment	Answers oral questions confidently Copies class work Shows understanding of the topic
5 mins	**EXTEND**	Tells learners to observe birds in their locality and identify which group they belong to Asks them to find out about reptiles for the next lesson Encourages them to read pages 22-23 of their source material Prepares them for next topic: Reptiles	Home assignment Preview next lesson	Listens to home assignment Asks questions where unclear Prepares for next lesson

CLASS WORK:

1. State any two ways in which birds are similar to mammals.
2. Why are sunbirds able to suck nectar from the bottom of flowers?
3. Give any one reason why birds are able to fly in air.
4. How is a humming bird different from other birds in the way it flies?

5. Why is an ostrich unable to fly?
6. How is the feeding of a pigeon different from that of a robin?
7. In which way is the feeding of a hornbill useful in the environment?
8. Identify the group of birds to which a turkey belongs.

LESSON EVALUATION:

STRENGTH:	AREAS WHICH NEED IMPROVEMENT:	WAY FORWARD:

LESSON PLAN 5: REPTILES

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Six	Integrated Science		60 Minutes

THEME	TOPIC	CONTENT
The World of Living Things	Classification of Animals	Reptiles: characteristics, examples, and adaptations

COMPETENCES:

- **Subject Competence:** The learner identifies reptiles, states their characteristics, and explains adaptations of different reptiles.
- **Language Competence:** The learner uses terms like cold-blooded, scales, moulting, venomous, constrictors, and fangs correctly.

TEACHING METHODS/TECHNIQUES:

- Question and answer
- Demonstration
- Guided discovery
- Use of charts/pictures
- Discussion
- Group work
- Note-taking

LEARNING OUTCOMES:

By the end of this lesson, learners should be able to:

1. State the characteristics of reptiles
2. Name examples of reptiles
3. Distinguish between poisonous and non-poisonous snakes
4. Explain adaptations of different reptiles to their environment

TEACHING/LEARNING MATERIALS:

- Chart showing different reptiles (refer to images on pages 23-25)
- Pictures of snakes (poisonous and non-poisonous), lizards, crocodiles, turtles, chameleons
- Chalkboard/whiteboard
- Learners' notebooks
- The P.6 Integrated Science Lesson Notes (Source document)

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	Greets learners and reviews groups of vertebrates Asks: "Name the five groups of vertebrates" (MR FAB) Shows pictures of a snake, lizard, crocodile, and turtle (pages 23-25) Asks: "What are these animals?" "What do they have in common?" Listens to learners' responses Introduces the topic: Reptiles	Question and answer Review Use of pictures Brainstorming	Responds to greeting Answers review questions Observes the pictures Names the animals seen Mentions features they share (scales, crawl) Listens attentively
15 mins	**EXPLORE**	Divides learners into groups Gives each group pictures of different reptiles Asks them to observe and list features they see (body covering, legs/no legs, where they live) Guides them to discuss: "How do these animals move?" "What covers their bodies?" Lets groups present their findings Records their observations on the board	Group work Guided discovery Observation Discussion	Works cooperatively in groups Observes reptile pictures carefully Lists features of reptiles Discusses movement and habitat Presents findings to class Listens to other groups
20 mins	**EXPLAIN**	Defines reptiles: cold-blooded vertebrates whose body is covered with scales and move by crawling (page 22) States characteristics of reptiles (page 22): • Their body is covered with scales • They breathe through lungs • They are cold-blooded • They undergo moulting • They reproduce by laying eggs • They undergo internal fertilization	Lecture method Explanation Use of pictures Drawing Note-taking	Listens attentively Observes pictures of reptiles Identifies characteristics of each reptile Writes notes in their books Answers questions Asks for clarification

		Lists examples of reptiles: snakes, lizards, crocodiles, turtles, chameleon, tortoise, terrapin Explains about snakes in detail (pages 22-23): • Characteristics: limbless; move by gliding/slithering; forked tongue for smelling and touch; moult to grow • Groups of snakes (page 23): - Poisonous/venomous snakes: have fangs, triangular head, poison glands; leave two fang marks; examples: cobra, puff adder, black mamba, gaboon viper (refer to page 23 images) - Non-poisonous snakes: have solid teeth, round head, no venom glands; leave four marks; examples: green grass snake, brown snake, house snake (refer to page 24 images) - Constrictors: kill prey by coiling around them; non-poisonous; lick prey to make it slippery; examples: python, anaconda, boa Draws and explains fangs and forked tongue (page 23) Explains adaptations of other reptiles (page 24): **Chameleon:** bulging eyes (see all directions); changes colour for protection; long sticky tongue to trap prey; well-developed feet and tail for gripping branches; feeds on harmful insects **Gecko:** sticky tiny hair under feet to grip walls/ceilings; feeds on harmful insects **Tortoises/Turtles:** protect themselves by hiding in hard shells; turtles have flippers for swimming **Crocodiles:** long strong tails and teeth for protection; live in water and on land; lay eggs in sand for warmth and protection Refers to diagrams on pages 23-25 Asks learners to copy notes		
10 mins	**ELABORATE**	Shows pictures of different snakes and asks learners to identify if they are poisonous or non-poisonous Asks questions: • "How can you identify a poisonous	Guided practice Question and answer Application Discussion	Identifies poisonous and non-poisonous snakes Explains adaptations of constrictors

		snake?" • "Why do pythons squeeze their prey?" • "Why do they lick their prey?" Guides learners to answer questions from the source material (pages 25-26, questions 1-8) Discusses importance of reptiles (page 25): • Source of food • Provide skins for leather • Eat crop pests • Control vectors • Venom used to make serum Discusses similarities and differences between reptiles and birds (page 25)		Answers questions from source material Participates in discussion Writes answers in their books
5 mins	**EVALUATE**	Asks oral questions: • "What are reptiles?" • "Name three characteristics of reptiles" • "How can you identify a poisonous snake?" • "What are constrictors? Give two examples" • "How is a chameleon adapted to catching insects?" Gives class work from the source material (pages 25-26, questions 9-12)	Oral assessment Question and answer Written assignment	Answers oral questions confidently Copies class work Shows understanding of the topic
5 mins	**EXTEND**	Tells learners to find out about fish for the next lesson Asks them to observe fish in water and note how they move and breathe Encourages them to read pages 26-27 of their source material Prepares them for next topic: Fish	Home assignment Preview next lesson	Listens to home assignment Asks questions where unclear Prepares for next lesson

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CLASS WORK:

1. Which group of cold-blooded vertebrates undergo internal fertilization?
2. State any two features used to identify poisonous snakes.
3. What name is given to the long teeth of a cobra?
4. Apart from cobra, give any two other examples of venomous snakes.
5. State the difference in structure between the teeth of a puff adder and those of a brown snake.
6. Why are fangs of a cobra positioned pointing backwards?
7. How are fangs of a gaboon viper able to inject poison into an enemy?
8. Give the reason why a python squeezes its prey.

9. How is a snake useful to a crop farmer?

LESSON EVALUATION:

STRENGTH:	AREAS WHICH NEED IMPROVEMENT:	WAY FORWARD:

LESSON PLAN 6: FISH

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Six	Integrated Science		60 Minutes

THEME	TOPIC	CONTENT
The World of Living Things	Classification of Animals	Fish: characteristics, external features, and adaptations

COMPETENCES:

- **Subject Competence:** The learner identifies fish, states their characteristics, and explains the functions of their external features.
- **Language Competence:** The learner uses terms like gills, fins, lateral line, scales, streamlined body, and swim bladder correctly.

TEACHING METHODS/TECHNIQUES:

- Question and answer
- Demonstration
- Guided discovery
- Use of charts/pictures
- Discussion
- Group work
- Note-taking

LEARNING OUTCOMES:

By the end of this lesson, learners should be able to:

1. State the characteristics of fish
2. Identify the external parts of a fish
3. Explain the functions of each part of a fish
4. State the importance of fish and why fish live in water

TEACHING/LEARNING MATERIALS:

- Chart showing a fish with labeled parts (refer to image on page 27)
- Picture of a fish (tilapia, Nile perch) from page 26

- Diagram of swim bladder (page 28)
- Real specimen (if available: fresh fish)
- Chalkboard/whiteboard
- Learners' notebooks
- The P.6 Integrated Science Lesson Notes (Source document)

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	Greets learners and reviews reptiles from previous lesson Asks: "Name the five groups of vertebrates" Shows a picture of a fish (tilapia) from page 26 Asks: "What animal is this?" "Where does it live?" "How does it move in water?" Listens to learners' responses Introduces the topic: Fish	Question and answer Review Use of pictures Brainstorming	Responds to greeting Answers review questions Observes the picture Names the animal and its habitat Mentions how fish move in water Listens attentively
15 mins	**EXPLORE**	Divides learners into groups Gives each group a picture of a fish (or real specimen if available) from page 27 Asks them to observe and list all the external parts they can see Guides them to identify: mouth, eyes, nostrils, gills, fins, scales, tail Asks: "What do you think each part is used for?" Lets groups present their findings Records their observations on the board	Group work Guided discovery Observation Discussion	Works cooperatively in groups Observes the fish picture carefully Lists external parts of fish Discusses possible functions of each part Presents findings to class Listens to other groups
20 mins	**EXPLAIN**	States characteristics of fish (page 26): • They are cold-blooded vertebrates • They entirely live in water • They reproduce by laying eggs • They undergo external fertilization • They have a slippery body to escape from predators • They have fins for swimming in water • They have streamlined bodies to reduce friction during swimming • They breathe through gills Draws and labels the external parts of a fish on the board (as on page 27)	Lecture method Explanation Drawing Demonstration Note-taking	Listens attentively Observes the diagram Identifies each part as teacher explains Writes notes and draws the diagram Asks questions Participates in discussion

		Explains functions of each part (pages 27-28): **Mouth:** For feeding; takes in water to extract dissolved oxygen **Nostril/nare:** For smelling food (works like feelers of insect, human nose) **Lateral line:** Detects sound vibrations in water; senses danger (works like human ear) **Dorsal fin:** For protection; has sharp spines; balances the fish; prevents rolling **Pelvic and pectoral fins:** For swimming upwards and downwards; increase or decrease speed (work like gears) **Caudal/tail fin:** Changes direction and moves forward; strong muscles for turning (works like steering wheel) **Gills:** For breathing (work like lungs) **Gill cover/operculum:** Protects gills (works like rib cage) **Scales:** Protect body; slippery to slip off predators; arranged to overlap to reduce friction Refers to diagram on page 27 Explains why fish are able to live in water (page 28): • Have gills for breathing • Take in dissolved oxygen Explains why fish dies when removed from water (page 26): • Lacks dissolved oxygen • Muscular system of mouth and gills cannot work in air Notes: fish has a swim bladder to keep it buoyant in water (page 28 diagram) Asks learners to copy diagram and notes		
10 mins	**ELABORATE**	Asks learners to label parts of a fish on a diagram Gives questions: • "How is the lateral line useful to a fish?" • "Why do fish have overlapping scales?" • "What is the function of the tail fin?" Guides learners to answer questions from the source material (page 28, questions 3-6)	Guided practice Question and answer Application Discussion	Labels parts of a fish Answers questions correctly Writes answers in their books Participates in discussion States importance of fish

		Discusses importance of fish (page 28): • Source of food • Source of income • Fish bones used for animal feeds and glue • Fish farming provides employment • Feed on harmful insects like mosquitoes Discusses similarities and differences between fish and reptiles (page 28)		
5 mins	**EVALUATE**	Asks oral questions: • "What are the characteristics of fish?" • "Name three fins of a fish and their functions" • "How does a fish breathe?" • "Why does a fish die when removed from water?" • "What is the function of the lateral line?" Gives class work from the source material (page 28, questions 1-6)	Oral assessment Question and answer Written assignment	Answers oral questions confidently Copies class work Shows understanding of the topic
5 mins	**EXTEND**	Tells learners to find out about amphibians for the next lesson Asks them to observe frogs or toads around their home and note their features Encourages them to read pages 29-30 of their source material Prepares them for next topic: Amphibians	Home assignment Preview next lesson	Listens to home assignment Asks questions where unclear Prepares for next lesson

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CLASS WORK:

1. To which group of vertebrates does a shark belong?
2. State any two characteristics of fish.
3. How are the following features useful to a fish?
   i) Streamlined body
   ii) Slippery body
4. Give any one reason why fish are able to live in water.
5. Why does a fish die soon after it has been removed from water?
6. Refer to the diagram on page 28: Name the parts marked P and U. Which structures are protected by part marked Q? Identify the part on the fish that has the same function as the human ear.

LESSON EVALUATION:

STRENGTH:	AREAS WHICH NEED IMPROVEMENT:	WAY FORWARD:

INCOMPLETE DOCUMENT. CALL OR WHATSAPP 0750-470-900 TO PURCHASE FULL COPY AND MORE FROM P.1-P.7 ALL SUBJECTS

SIGN: Mr. Jayson